

tximeta support for mixed reference sets

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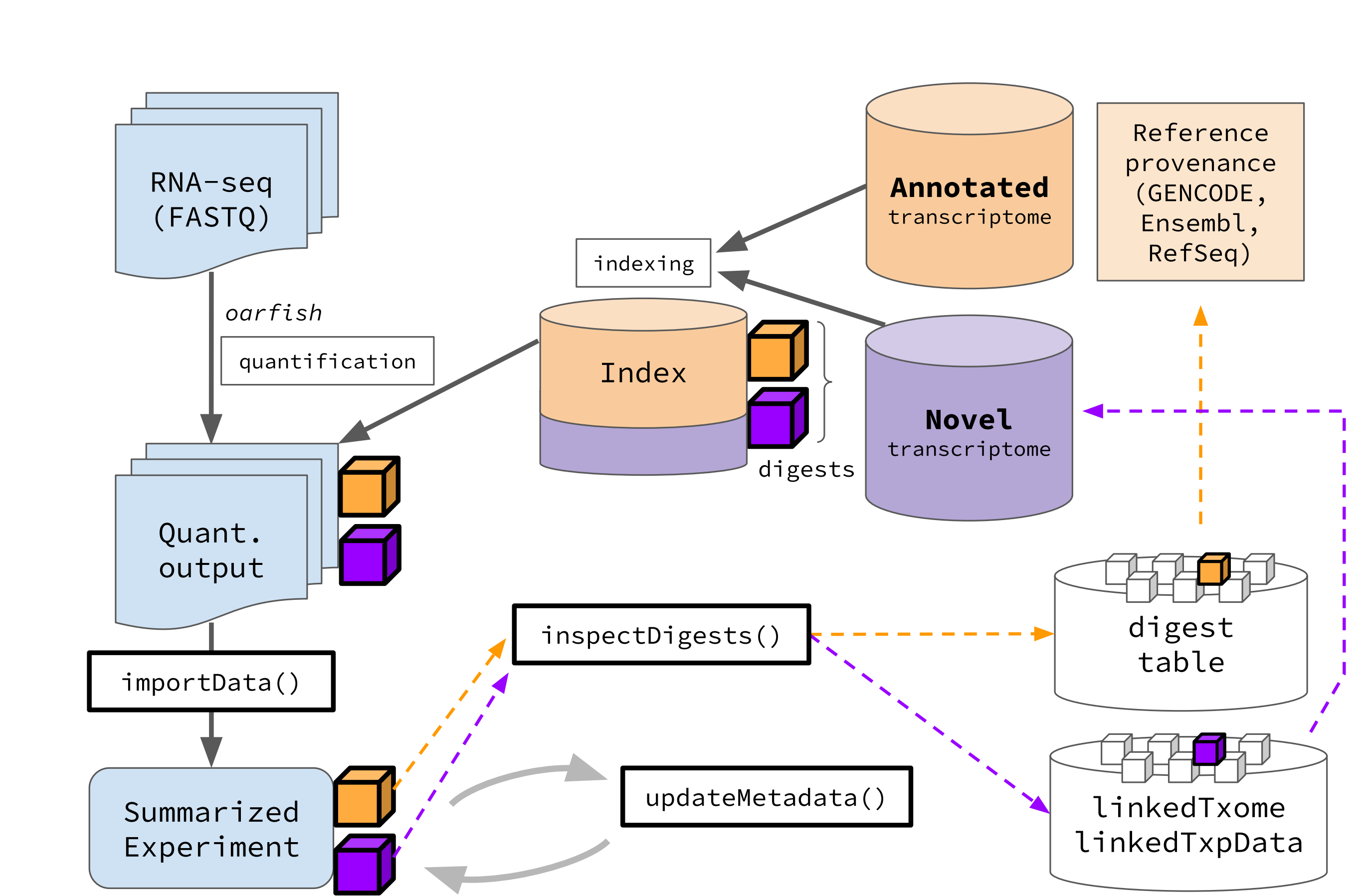


Figure 1: *tximeta* new workflow for mixed reference transcripts using annotated and novel digests.

Introduction

RNA-seq quantification tools such as *salmon* and *oarfish* operate against an index built from reference transcript sequences. A key feature of these tools is that they record a digest of those sequences at index-build time. *tximeta* reads that digest at import time to automatically identify the provenance of the reference. It then attaches the appropriate transcript and genome metadata to the resulting *SummarizedExperiment*, in the `rowData` and `rowRanges`. This workflow eliminates the need for users to manually track which annotation release was used, and to manually reconstruct metadata tables. It works regardless of the actions of the initial analyst performing quantification.

The long-read quantifier *oarfish* and *tximeta* recently extended this model to *mixed reference* indexing: a single index may combine a set of annotated transcripts (e.g. from GENCODE or Ensembl) with a set of novel transcripts (e.g. *de novo* assembled, spike-in, or fusion sequences not present in the reference annotation).

tximeta v1.28 (Bioc 3.22, October 2024) introduced the following functions, `importData()` → `inspectDigests()` → `updateMetadata()`, allowing the user to check each sub-index digest independently, add linked metadata, and populate transcript metadata wherever a match is found. Another new function, `linkedTxpData()` allows simple tabular metadata on transcripts to be stored in the file cache, where previously only GTF-based metadata was supported in *tximeta* via `linkedTxome()`.

Setup

Three replicates from the SGNEx H9 cell line, quantified with *oarfish* against a mixed reference: GENCODE v48 annotated transcripts plus a set of novel transcripts.

```
library(tximeta)
library(SummarizedExperiment)
files <- file.path(dir, paste0("sgnex_h9_", names, ".quant.gz"))
coldata <- data.frame(files, names)
```

Import

`importData()` reads the counts and stores the index digests in metadata, returning an un-ranged *SummarizedExperiment*.

```
se <- importData(coldata, type = "oarfish")

reading in files with read_tsv

1 2 3
returning un-ranged SummarizedExperiment, see functions:
-- inspectDigests() to check matching digests
-- makeLinkedTxome/makeLinkedTxpData() to link digests to metadata
-- updateMetadata() to update metadata and optionally add ranges
```

Inspect digests

`inspectDigests()` checks each index digest against *tximeta*'s registries. The annotated index matches GENCODE v48 in the pre-computed hash table; the novel index has no match yet.

```
inspectDigests(se)

# A tibble: 2 × 8
  index source organism release genome linkedTxome linkedTxpData small_digest
<chr> <chr> <chr> <chr> <chr> <lgl> <lgl> <chr>
1 annotated LocalGENCODE Homo sapiens 48 GRCh38 TRUE FALSE 6fc626
2 novel <NA> <NA> <NA> <NA> NA NA 43158f
```

Update metadata

`updateMetadata()` populates `rowData` for any transcripts whose index has a known match. To also attach genomic ranges, use `ranges = TRUE`.

```
se_updated <- updateMetadata(se)
## --annotated: adding metadata for 385669 transcripts
## --novel: no transcript metadata found
## consider using `linkedTxome`, or `linkedTxpData` (see man pages)
## --novel: no matching transcripts
```

Supplying novel transcript metadata

Novel transcript metadata can be provided directly via the argument `txpData`, or one can use `linkedTxpData()` to register them persistently, using the file cache. `txpData` can take tabular data or *GRanges* as input, here we show with ranges.

```
library(GenomicRanges)
head(novel, 2)
## seqnames start end strand tx_name gene_id
## 1 chr1 1000001 1001000 + novel1 novel_gene1
## 2 chr1 1001001 1002000 + novel2 novel_gene1
novel_gr <- as(novel, "GRanges"); names(novel_gr) <- novel$tx_name

se_full <- updateMetadata(se, txpData = novel_gr, ranges = TRUE)
## --annotated: adding metadata for 385669 transcripts
## --novel: no transcript metadata found
## consider using `linkedTxome`, or `linkedTxpData` (see man pages)
## --novel: no matching transcripts
## --user: adding metadata for 11000 transcripts
## building RangedSE: subsetting to 396669 out of 423044 rows with range data
```

Persistent metadata via linking

Rather than supplying `txpData` each time, novel transcript metadata can be registered once in the file cache with `makeLinkedTxome` or `makeLinkedTxpData()`. The novel index digest is obtained from the `inspectDigests()` output or from the *oarfish* output directly.

```
makeLinkedTxpData(
  digest = "43158f2c8e88e3acd77c22aee557625a6f1b6a5038cfc7deb5e64903892d8070",
  digestType = "sha256", indexName = "my_novel_txps", txpData = novel_gr,
  source = "novel", organism = "Homo sapiens", release = "v1", genome = "GRCh38"
)
## linking user-provided metadata to digest: 43158f...
## saving txpData object in bfc
## saving linkedTxpData in bfc (first time)
```

Digests now recognized

After registration, `inspectDigests()` reflects the match.

```
inspectDigests(se)

# A tibble: 2 × 8
  index source organism release genome linkedTxome linkedTxpData small_digest
<chr> <chr> <chr> <chr> <chr> <lgl> <lgl> <chr>
1 annotated LocalGENCODE Homo sapiens 48 GRCh38 TRUE FALSE 6fc626
2 novel novel Homo sapiens v1 GRCh38 FALSE TRUE 43158f
```

`updateMetadata()` finds and applies the linked metadata.

```
se_linked <- updateMetadata(se)
## --annotated: adding metadata for 385669 transcripts
## --novel: adding metadata for 11000 transcripts
```

Next steps

Backporting to *salmon*. The mixed reference workflow currently targets *oarfish*, which natively records separate digests for each sub-index within the *minimap2* index itself, and in the quantification metadata JSON files (*.meta_info.json). *salmon* does not record per sub-index digests, so the new import model is not yet available to *salmon* users who combine reference transcripts. Extending support will require either a change to *salmon*'s metadata output, or a preprocessing step handle by a wrapper tool, or workflow rule, that records the sub-index digests into a standardized location alongside indexing.

Quantification over subsets of the annotated transcript set. A common use case is to quantify against a subset of the annotated reference, e.g. a tissue- or cell-type-specific transcriptome. Digest-based provenance breaks down here because the subset has a different hash from any entry in the pre-computed table, so *tximeta* cannot identify it automatically. Again, a wrapper tool or workflow rule may be a solution.

Other updates

New importer support for *edgeR* and *limma*. New function `edgeR::DGEListFromTximeta()` assists with downstream analyses with other Bioconductor packages, including dividing out count uncertainty for isoform-level analysis. See *tximeta* vignette.

New argument for GENCODE. `gencode_gtf_prefix` allows specifying the desired GTF from GENCODE, e.g. "chr_patch_hapl_scaff".

Ranges after summarization. `summarizeToGene()` argument `assignRanges` takes either "range" (default) or "abundant", for determining the ranges that are attached to the SE. Default "range" provides the range of all exons of the gene. "abundant" instead returns the range of the most abundant isoform in the data. See `?summarizedToGene`.

Feedback

Please provide feedback via: <https://github.com/thelovelab/tximeta/issues>

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